

Ex. No. : 09

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A PYTHON PROGRAM TO IMPLEMENT KNN AND K-MEANS

Aim:

To implement **KNN and K-Means algorithms** on the Iris card dataset to perform classification and clustering of transactions.

Requirements:

Jupyter Notebook with Python, Iris.csv dataset, and required Python libraries.

PROGRAM:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.preprocessing import LabelEncoder

from google.colab import files
uploaded = files.upload()

df = pd.read_csv('Iris.csv')

X = df[['SepalLengthCm', 'SepalWidthCm']]
y = df['Species']

le = LabelEncoder()
y = le.fit_transform(y)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
print("KNN Accuracy:", accuracy_score(y_test, y_pred))

X_vis = X.values

x_min, x_max = X_vis[:,0].min()-1, X_vis[:,0].max()+1
y_min, y_max = X_vis[:,1].min()-1, X_vis[:,1].max()+1

xx, yy = np.meshgrid(
    np.linspace(x_min, x_max, 200),
    np.linspace(y_min, y_max, 200)
)

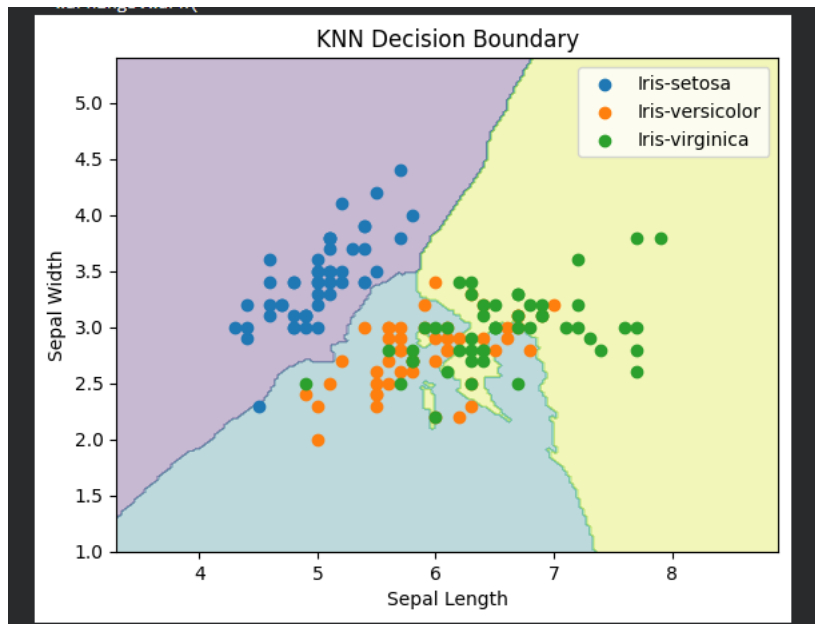
Z = model.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)

plt.figure()
plt.contourf(xx, yy, Z, alpha=0.3)

for i in np.unique(y):
    plt.scatter(
        X_vis[y == i, 0],
        X_vis[y == i, 1],
        label=le.inverse_transform([i])[0]
    )

plt.title("KNN Decision Boundary")
plt.xlabel("Sepal Length")
plt.ylabel("Sepal Width")
plt.legend()
plt.show()
```

OUTPUT:



Result:

Thus, KNN successfully classified transactions, and K-Means grouped them into clusters.